

Lecture 17 – Energy Storage

Part I: Why Energy Storage Matters

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Outline

- Energy Storage Overview
- Energy Storage Functionality
- Key Concepts of Storage
- Storage Scheduling

1. B2
2. Dave Mooney, NREL: *GCEP Tutorial*

Energy Storage

- Energy storage (accumulator/battery): capture of energy produced at one time for use at a later time.
- Storage shifts energy through time rather than creating energy
- Storage converts energy from forms that are difficult to store to more conveniently/economically storable forms.



Why Is Everyone Talking About Storage?

Rapid growth of solar and wind generation

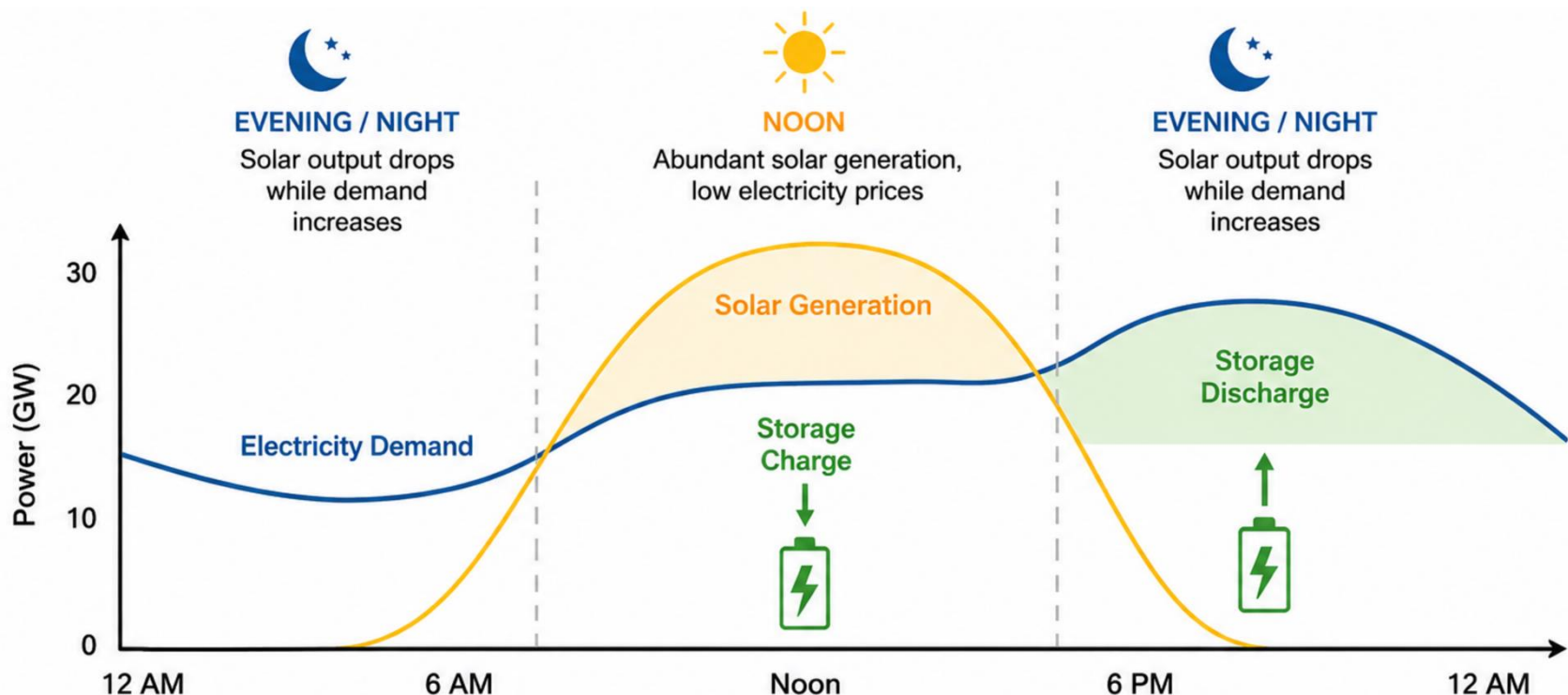
- Increasing EV adoption and electrification

- Growing reliability concerns and extreme weather events

- Storage is becoming a critical grid asset

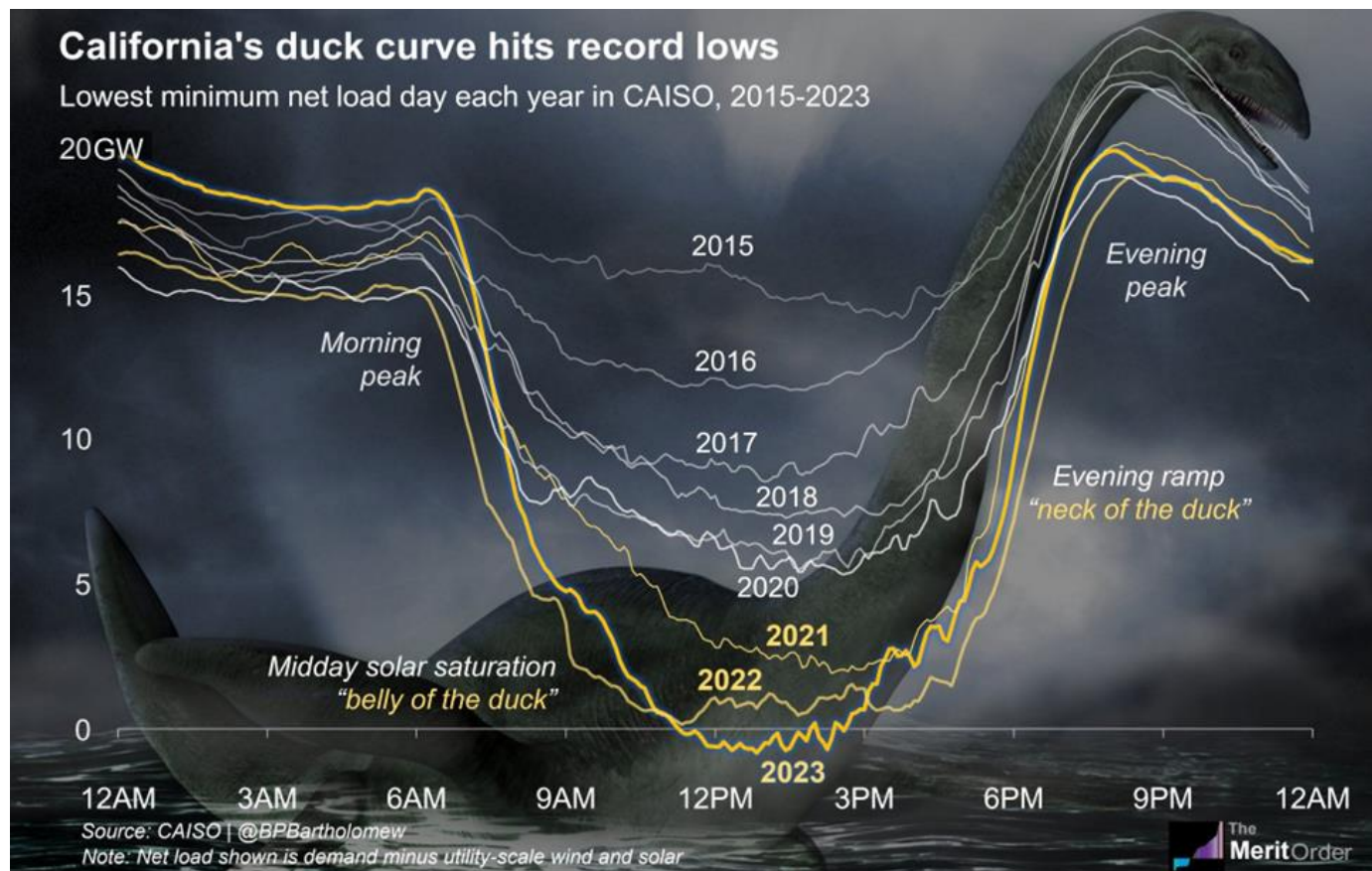
Why Does the Grid Need Energy Storage?

- Noon: abundant solar generation, low electricity prices
- Evening: solar output drops while demand increases
- Question: How do we supply power when solar is unavailable?
- Storage bridges the gap between generation and demand



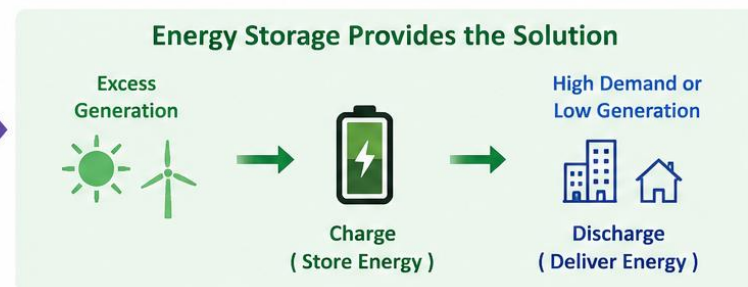
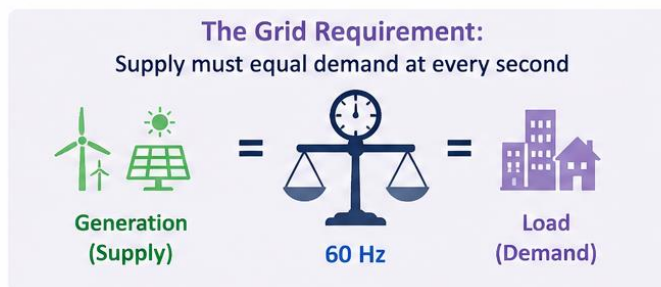
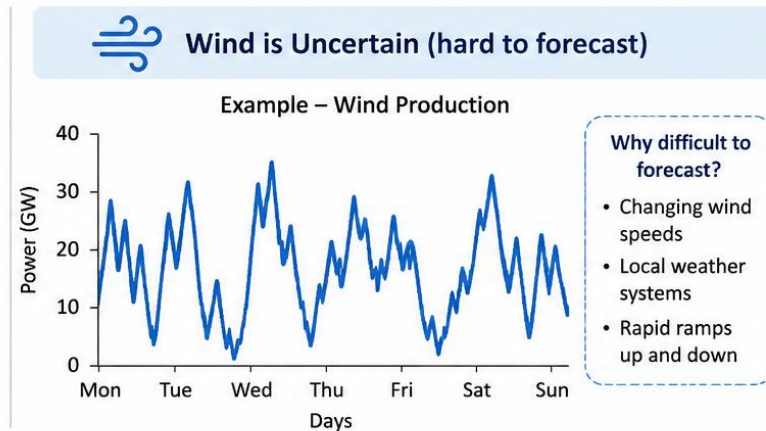
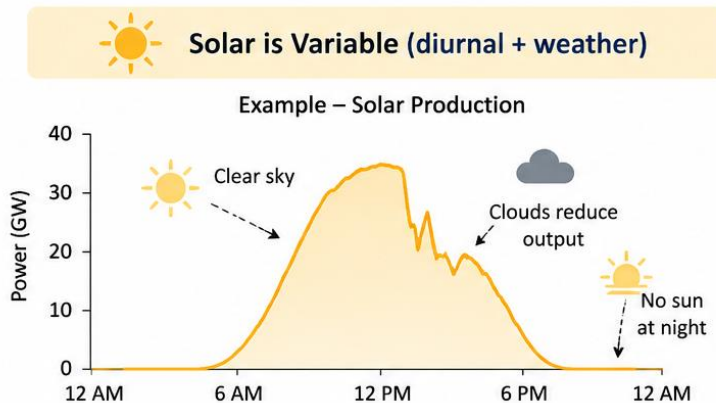
The California Duck Curve

- Net Load = Demand – Renewable Generation
- Solar reduces net load during midday
- Steep evening ramp after sunset
- Storage absorbs excess solar and discharges during peak demand



Renewable Energy Creates New Challenges

- Solar output varies with weather and time of day
- Wind generation is uncertain and difficult to forecast
- Supply and demand must be balanced every second
- Storage provides flexibility and reliability



Bottom Line: Renewables are clean and abundant, but variability and uncertainty create challenges.
Energy storage adds flexibility and reliability to keep the grid stable.

Grid Flexibility

The timing of energy consumption often does not align with the timing of energy production.

Some energy sources are intermittent:

- Biofuels (grass, sugar cane, corn, and oilseeds) are only available during part of the year.
- Solar, wind, and ocean motion energy sources have roughly daily cycles. They add **variability** and **uncertainty** to the generation supply, increasing the need for grid **flexibility**.

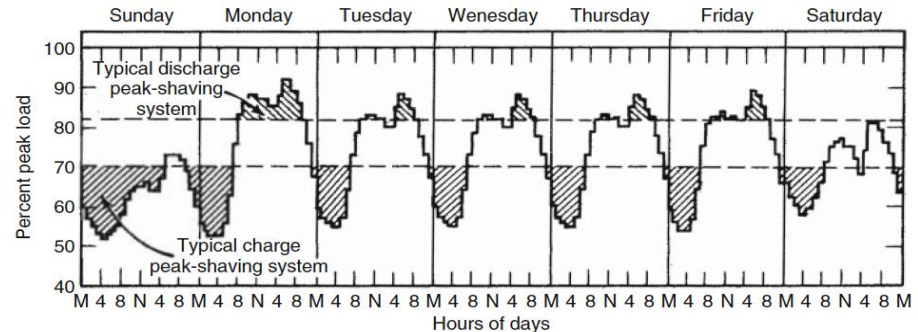
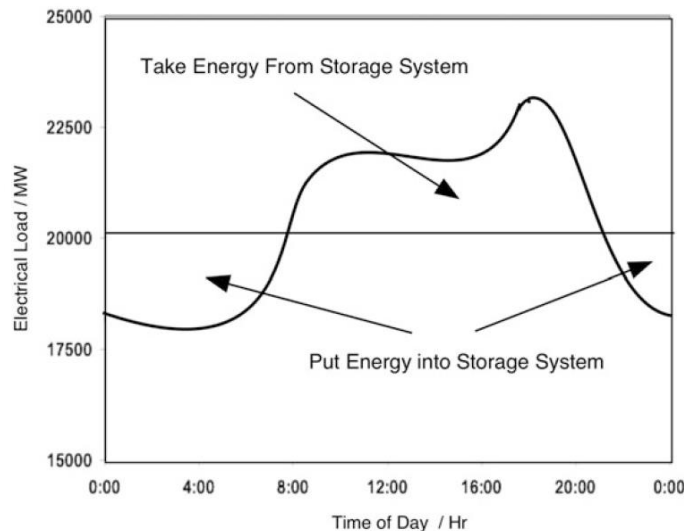
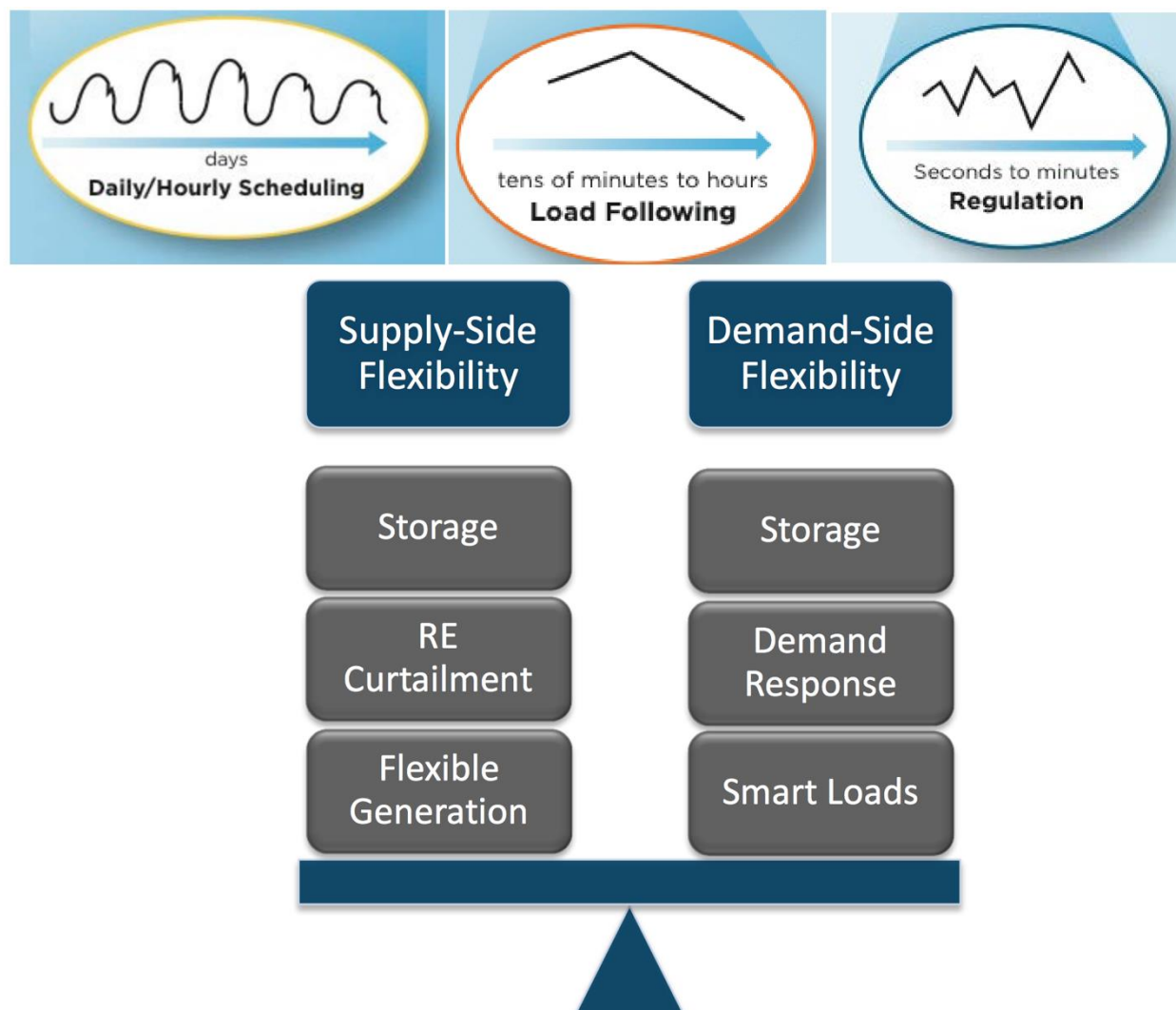


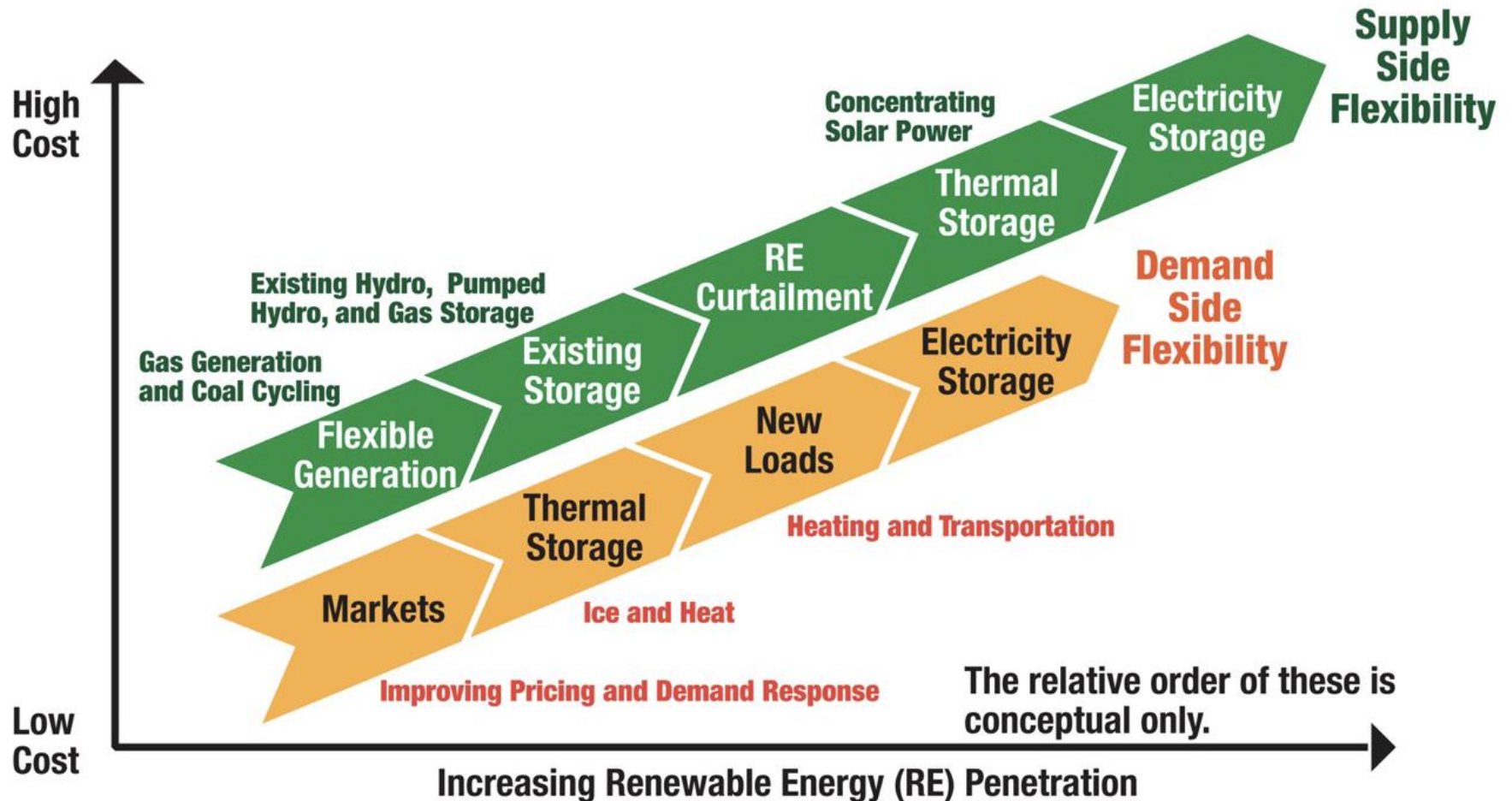
Fig. Hourly data on electric load show possible use of a storage system.

Flexibility to Maintain Balance



Storage is one of many options for providing more flexible supply and demand at multiple timescales.

Flexibility Supply Curve



Electricity Markets Operate Across Complementary Time Horizons

Energy storage provides flexibility and value across all time horizons

TIME HORIZON



DAYS

Forward planning



HOURS

Scheduling and dispatch



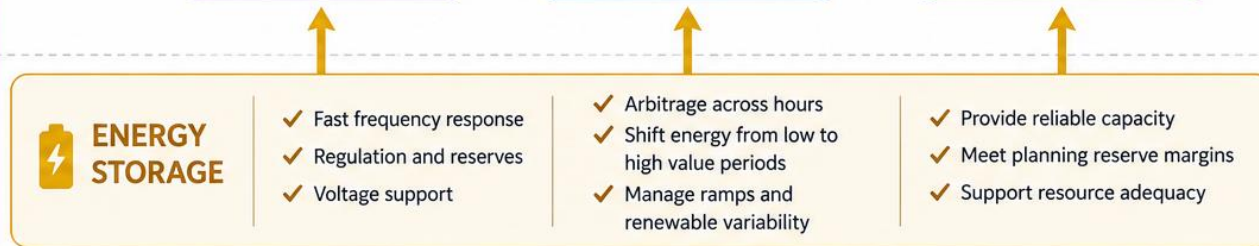
MINUTES

Real-time adjustments



SECONDS

Instantaneous stability



PURPOSE

Maintain System Stability

Act in seconds to minutes



Operate the System Efficiently

Optimize dispatch in real time to hours



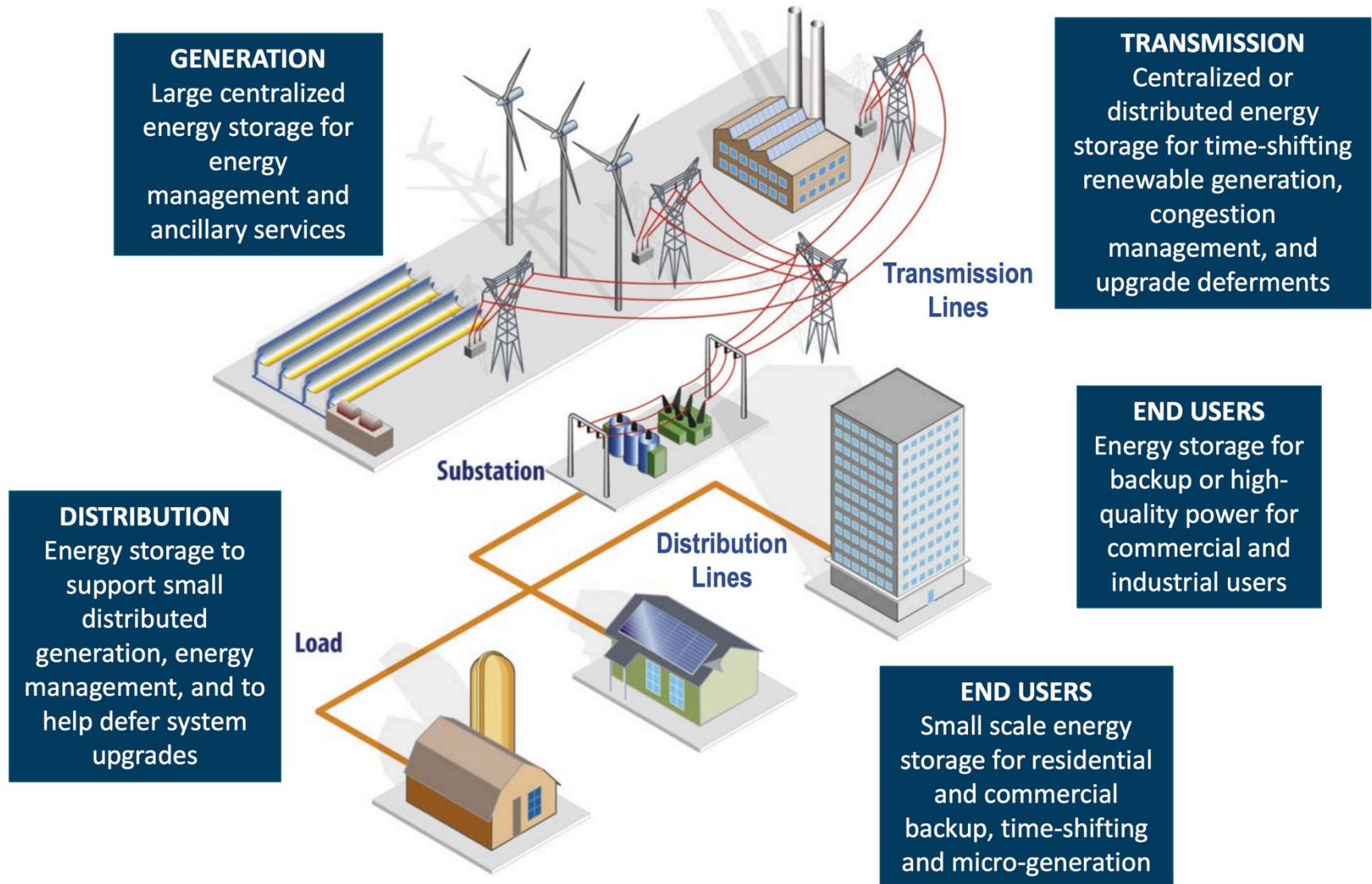
Plan for Adequacy

Ensure enough resources for the future



Energy storage is a flexible asset that enables markets at every time horizon—enhancing reliability, efficiency, and resilience of the power system.

Energy Storage Applications



Key Concepts of Energy Storage

1. Energy Capacity (MWh)
2. Power Rating (MW)
3. Discharge Duration (h)
4. Response Time (ms–min)
5. Round-Trip Efficiency (%)
6. State of Charge (SOC)
7. Depth of Discharge (DoD)
8. Cycle Life (# cycles)
9. Energy Density (Wh/kg, Wh/L)
10. Power Density (W/kg, W/L)

1. Energy Capacity

How much energy the storage system can hold.

- Similar to the size of a fuel tank in a car.
- Determines how long the storage system can supply electricity.
- Common units:
 - Wh (watt-hour)
 - kWh (kilowatt-hour)
 - MWh (megawatt-hour)

Examples:

- Smartphone battery: 5,000 mAh ($\approx$ 19 Wh)
 - Tesla Model Y battery: $\sim$ 75 kWh
 - Grid-scale battery: hundreds of MWh
-

2. Power Rating

How fast the storage system can deliver or absorb energy.

- Similar to the engine power of a car.
- Determines the maximum charging or discharging rate.

Units:

- W, kW, MW

Examples:

- A battery rated at **1 MW** can inject up to **1 MW** of power into the grid.
- A battery rated at **100 kW** can charge or discharge at a maximum rate of **100 kW**.

3. Discharge Duration

How long the storage system can operate at its rated power.

$$\text{Discharge Duration} = \frac{\text{Energy Capacity}}{\text{Power Rating}}$$

Example:

- Energy Capacity = 5 MWh
- Power Rating = 1 MW

$$\text{Duration} = 5 \text{ h}$$

4. Response Time

How quickly the storage system can change its output after receiving a control signal.

Typical values:

Technology	Response Time
Li-ion Battery	milliseconds to seconds
Flywheel	milliseconds
Gas Turbine	minutes
Coal Plant	tens of minutes to hours

Fast response is critical for frequency regulation and grid stability.

5. Round-Trip Efficiency (RTE)

The percentage of electrical energy recovered after a complete charge-discharge cycle.

$$\text{RTE} = \frac{\text{Energy Out}}{\text{Energy In}} \times 100\%$$

Approximately,

$$\text{RTE} \approx \eta_{\text{charge}} \times \eta_{\text{discharge}}$$

where:

- η_{charge} : charging efficiency
- $\eta_{\text{discharge}}$: discharging efficiency

Example:

- Charging efficiency = 95%
- Discharging efficiency = 95%

$$\text{RTE} = 0.95 \times 0.95 = 90.25\%$$

Higher RTE means less energy is lost during storage.

6. State of Charge (SOC)

The percentage of energy currently stored in the battery.

$$SOC = \frac{\text{Current Stored Energy}}{\text{Maximum Energy Capacity}} \times 100\%$$

- 100% SOC = fully charged
 - 50% SOC = half charged
 - 0% SOC = empty
-

7. Depth of Discharge (DoD)

The percentage of battery capacity that has been used.

$$DoD = 100\% - SOC$$

- 0% DoD = fully charged
- 100% DoD = fully discharged

Why it matters:

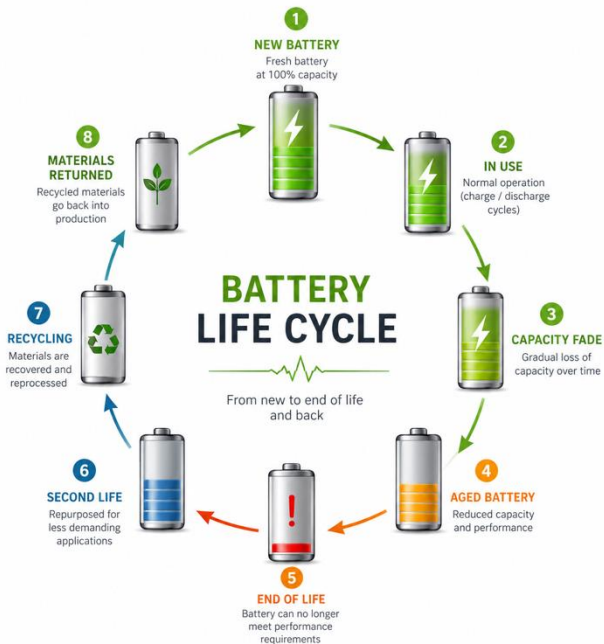
- Higher DoD provides more usable energy.
- Frequent deep discharges generally shorten battery life.

8. Cycle Life

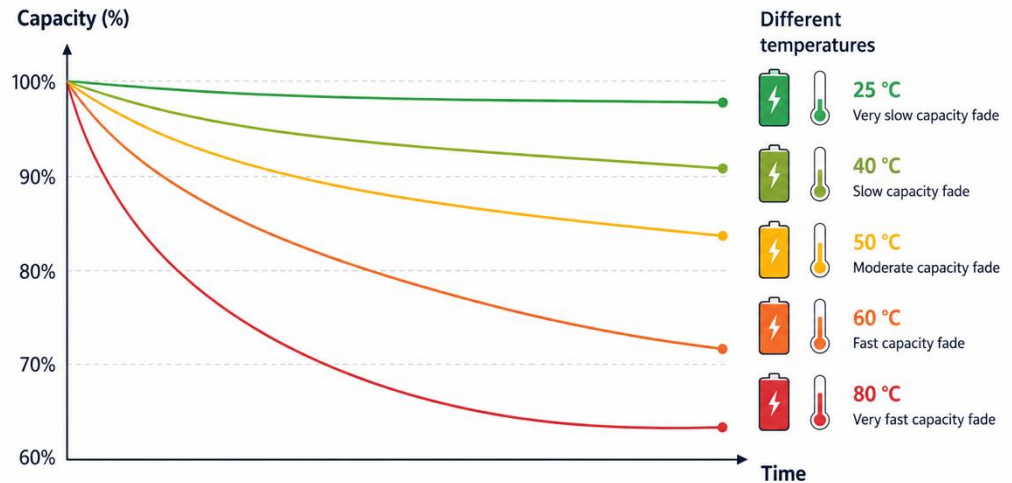
The number of charge-discharge cycles a battery can undergo before significant degradation occurs.

Typical values:

Technology	Cycle Life
Lead-acid	500–1,500
Li-ion	3,000–10,000+
Flow Battery	10,000–20,000+



Battery Capacity Fades Faster at Higher Temperatures



Higher temperatures accelerate battery aging and reduce usable capacity over time. Keeping batteries cool extends their life and maintains performance.

9. Energy Density

The amount of energy stored per unit mass or volume.

Energy density indicates how much energy a storage system can hold relative to its size or weight.

$$\text{Gravimetric Energy Density} = \frac{\text{Stored Energy (Wh)}}{\text{Mass (kg)}}$$

$$\text{Volumetric Energy Density} = \frac{\text{Stored Energy (Wh)}}{\text{Volume (L)}}$$

Units:

- Wh/kg (gravimetric)
- Wh/L (volumetric)

Examples:

Technology	Energy Density (Wh/kg)
Lead-acid Battery	30–50
Lithium-ion Battery	150–300
Gasoline	~12,000*
Pumped Hydro	Very Low

*Gasoline stores chemical energy and is not directly comparable to electrical storage.

Why it matters:

- Determines weight and size.
- Critical for electric vehicles, drones, and portable electronics.
- Less important for large stationary grid storage.

10. Power Density

The amount of power delivered per unit mass or volume.

Power density indicates how quickly stored energy can be released.

$$\text{Gravimetric Power Density} = \frac{\text{Power Output (W)}}{\text{Mass (kg)}}$$

$$\text{Volumetric Power Density} = \frac{\text{Power Output (W)}}{\text{Volume (L)}}$$

Units:

- W/kg
- W/L

Examples:

Technology	Power Density
Supercapacitor	Very High
Flywheel	High
Lithium-ion Battery	Moderate
Pumped Hydro	Low to Moderate

Why it matters:

- Determines how fast energy can be delivered.
- Important for frequency regulation and acceleration.
- High power density enables rapid charge/discharge.

Energy Density vs. Power Density

A useful analogy:

Metric	Analogy
Energy Density	Size of a water tank
Power Density	Diameter of the outlet pipe

- A **large tank with a small pipe** → high energy density, low power density.
- A **small tank with a large pipe** → low energy density, high power density.

Typical Technology Comparison

Technology	Energy Density	Power Density
Pumped Hydro	Low	Low–Moderate
Lithium-ion Battery	High	Moderate–High
Flow Battery	Moderate	Moderate
Flywheel	Low	High
Supercapacitor	Very Low	Very High

SOC Dynamics: Two-Variable Formulation

Method 1: Separate Charging and Discharging Variables

State Equation

$$E_{k+1} = E_k + \eta_c P_{c,k} \Delta t - \frac{P_{d,k}}{\eta_d} \Delta t$$

or equivalently

$$SOC_{k+1} = SOC_k + \frac{\eta_c P_{c,k} \Delta t}{E_{\max}} - \frac{P_{d,k} \Delta t}{\eta_d E_{\max}}$$

where

- P_c : charging power (kW)
- P_d : discharging power (kW)
- η_c, η_d : charging/discharging efficiencies
- $E_{\max}$: battery capacity (kWh)

Complementarity Constraint

A battery inverter is usually operating in one of three modes:

Mode	Charging Power	Discharging Power
Charging	> 0	0
Idle	0	0
Discharging	0	> 0

At the DC battery terminals, the battery cannot simultaneously absorb and deliver power.

Therefore, from a physical modeling perspective,

$$P_c P_d = 0$$

SOC Dynamics: Single-Variable Formulation

Define Battery Power

$$P_b > 0 : \text{Discharging} \quad P_b < 0 : \text{Charging} \quad P_b = 0 : \text{Idle}$$

where P_b is the battery power (kW).

Battery Energy Dynamics

Let

- E_k : stored energy at time step k (kWh)
- $E_{\max}$: battery capacity (kWh)
- η_c : charging efficiency
- η_d : discharging efficiency
- Δt : time interval (h)

Then

$$E_{k+1} = E_k + \begin{cases} -\frac{P_{b,k}\Delta t}{\eta_d}, & P_{b,k} > 0 \\ -\eta_c P_{b,k}\Delta t, & P_{b,k} < 0 \end{cases}$$

SOC Dynamics

Since

$$SOC_k = \frac{E_k}{E_{\max}},$$

the SOC evolves according to

$$SOC_{k+1} = SOC_k + \begin{cases} -\frac{P_{b,k}\Delta t}{\eta_d E_{\max}}, & P_{b,k} > 0 \\ -\frac{\eta_c P_{b,k}\Delta t}{E_{\max}}, & P_{b,k} < 0 \end{cases}$$

Battery Scheduling = Optimal Control

State:

$$SOC(t)$$

Control:

$$P_b(t)$$

Dynamics:

$$SOC_{k+1} = f(SOC_k, P_{b,k})$$

Objective:

$$\min \text{Cost}$$

Subject to:

$$SOC_{min} \leq SOC \leq SOC_{max}$$